Temperature at inactive black holes Within the Universal Quantum Foam Hypothesis (UQSH)

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Conceptual Origin November 8, 2025
March 01, 2026 on Figshare
January 25, 2026 on Zenodo

https://doi.org/10.6084/m9.figshare.31441018

Abstract

Inactive black holes should have effective temperatures below the cosmic microwave background radiation of 2.7 K, with this temperature decreasing further with increasing mass. This assumption is not understood here as an exotic property of an object, but rather as an indication of a special state of space itself. Within the framework of the Universal Quantum Foam Hypothesis, black holes appear as zones of maximum field saturation, in which open dynamics do not disappear but transform into internal stress forms. Temperature thus becomes a measure of the mobility of the medium. The unusual coldness of inactive black holes, as postulated by Hawking, therefore does not point to emptiness, but to an extremely bound field state. The comparison with the cosmic microwave background radiation opens up a new approach to observing these states and investigating the mechanical properties of space itself.

1 Temperature at inactive black holes

Within the framework of the Universal Quantum Foam Hypothesis, the temperature of a black hole takes on a fundamentally new meaning. It is not understood as an independent thermodynamic quantity but rather as a macroscopic signature of the field mechanical state of the medium. In this model, temperature does not primarily measure heat in the classical sense but the degree of internal mobility and reorganization capacity of the quantum foam.

An inactive black hole that exhibits neither significant accretion nor pronounced emission resides in a state of maximal field saturation. The available dynamics of the medium are largely converted into internal stress organization. The quantum foam is so highly compressed that scarcely any space remains for free oscillation, open reorganization, or propagating motion. The dynamics that appear as thermal activity in less saturated regions are not eliminated but are almost entirely bound into internal cycles.

This leads to a characteristic consequence. While even intergalactic space exhibits a minimal excitation level of about 2.7 K due to the cosmic microwave background radiation, this microscopic activity is structurally suppressed in the region of an inactive black hole. The object does not appear cold because it lacks energy but because the extreme field tension inhibits any form of open oscillation from which temperature arises in the first place. In the logic of the UQSH, an inactive black hole is thus a thermal sink whose effective temperature lies below the already low level of its surroundings.

A small inactive black hole reaches saturation within a spatially confined region. Although the field structure there is extremely taut, the transition between the saturated and free medium remains steep. A small amount of residual dynamics may persist in this transition zone and should manifest as a relatively higher effective temperature, although still well below 2.7 K. A large black hole distributes the same saturation state over a significantly greater volume. The transition to the free quantum foam becomes shallower, the field structure more homogeneous and mechanically calm. Gradients, microscopic friction, and opportunities for residual motion diminish. The larger the black hole, the closer the field state approaches near complete internalization of dynamics. Accordingly, the effective temperature decreases. A very small black hole would be warmer in this sense because its field state remains comparatively turbulent. A supermassive black hole would appear extremely cold, closer to the minimal possible openness of the medium than any other known object.

This field mechanical interpretation does not contradict established theoretical predictions. The Hawking temperature, derived from quantum field theory, likewise predicts extremely low temperatures for massive black holes, far below the level of the cosmic microwave background radiation. Such temperatures, however, have not yet been measured experimentally, as they are practically inaccessible to existing instruments.

This perspective opens a new observational approach. Through highly precise temperature comparisons between black holes and the level of the cosmic microwave background radiation, the degree of field saturation in different gravitational environments could be inferred, not as a direct measurement of Hawking radiation but as an indirect probe of the mechanical properties of the medium itself. Such measurements would be technically formidable, yet they could bridge the gap between cosmological field mechanics and observable physics.

It is essential to distinguish between the environment of a black hole and its internal field state. Accretion can generate an extremely hot surrounding region in the form of luminous accretion disks, jets, and high energy radiation. This thermal activity, however, belongs to the external dynamics of the medium and not to the internal state of the black hole itself.

Within the logic of the UQSH, the saturated core region remains structurally oriented toward maximal internalization. A black hole does not become "warmer" in a field mechanical sense through accretion. Incoming energy is immediately converted into internal stress organization. The temperature of the black hole itself therefore remains lower than that of its environment, regardless of how hot that environment appears. Active black holes illustrate this contrast most clearly: an extremely cold, almost motionless field core is surrounded by a highly dynamic, radiant envelope.

The observed heat is an expression of external field reorganization, not of internal mobility. Inactive black holes merely make this condition visible, because the outer envelope is absent and the intrinsically cold field state appears directly.

Parts of the editorial preparation were carried out with the assistance of AI-based tools.

Citation

Rexhepi, U. Q. (2026). *Temperature at inactive black holes Within the Universal Quantum Foam Hypothesis (UQSH)* .

Figshare: https://doi.org/10.6084/m9.figshare.31441018

Book on Amazon KDP: https://www.amazon.de/dp/B0GPQMX92G